

Decent Coaching Classes
IX CBSE
Mind Your Math - VI

Subject : Mathematics (Ch. 1 to 7)

Marks : 30

Note : All questions are compulsory. Section C has internal choice in each question.

Section A — Multiple Choice Questions (1 mark each)

1. The point $(-3, 4)$ lies in which quadrant? [1]
(a) I (b) II (c) III (d) IV
2. The degree of the polynomial $5x - 7$ is : [1]
(a) 0 (b) 1 (c) 2 (d) 5
3. Which of the following is an irrational number? [1]
(a) $4/9$ (b) $\sqrt{16}$ (c) $\sqrt{7}$ (d) $0.333\dots$
4. The value of $(a + b)^2 - (a - b)^2$ is : [1]
(a) $2ab$ (b) $4ab$ (c) $2a^2$ (d) $2b^2$
5. The angle subtended by a diameter at any point on the circle is : [1]
(a) 45° (b) 60° (c) 90° (d) 180°
6. The area of a triangle with base 10 cm and height 6 cm is : [1]
(a) 60 cm^2 (b) 30 cm^2 (c) 16 cm^2 (d) 120 cm^2

Section B — Very Short Answer Questions (2 marks each)

7. Write the coordinates of a point that lies on the y-axis, 5 units above the origin. [2]
8. Find the zero of the polynomial $p(x) = 3x - 12$. [2]
9. Insert two rational numbers between $1/4$ and $1/2$. [2]
10. A coin is tossed 40 times, and heads come up 22 times. Find the experimental probability of getting a tail. [2]

Section C — Short Answer Questions (4 marks each, with internal choice)

11. Simplify : $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$ [4]

OR

Using a suitable identity, evaluate : 97^2

12. In a circle with centre O, chord $AB = 8 \text{ cm}$ and the radius is 5 cm . Find the distance of the chord from the centre. [4]

OR

A chord of a circle is 10 cm long and its distance from the centre is 12 cm . Find the radius of the circle.

13. Find the area of a triangle whose sides are 13 cm , 14 cm and 15 cm , using Heron's formula. [4]

OR

Find the perimeter and area of a rectangular park whose length is 24 m and breadth is 18 m .

Section D — Case Study Based Question (4 marks)

14. A bag contains 5 red balls, 3 blue balls and 2 green balls. A ball is drawn at random from the bag. Based on this, answer the following :

- (a) Find the probability of drawing a red ball. [2]

(b) Find the probability of drawing a ball that is not green. [1]

(c) If the experiment is repeated 100 times, how many times would you expect a blue ball to appear (based on theoretical probability)? [1]